Naive Bayes ...

α - Laplaplace Smoothing.

α - hyperparameter, 'Tuning'

$\boxed{\alpha \text{ - impacts}}$

large value of 'α' ?

$$P(w_j \mid spam) \simeq P(w_j \mid ham)$$

likelihood.

$$P(w_j \mid y=1) \qquad P(w_j \mid y=0)$$

$$P(w_j \mid y=1) = \frac{n_1 + \alpha}{n_1 + 2\cdot\alpha} \qquad \boxed{C=2}$$

α - small

$$= \frac{90+1}{100+2\cdot 1} = \frac{91}{102} \qquad \alpha = 1$$

~ 0.9

$$\underbrace{p(sp \mid w)}_{} \; \boxed{\overset{0.5}{P(spam)} \cdot \overset{0.9}{P(w_j \mid spam)}} \quad \checkmark \qquad P(w_j \mid spam) \,\Big|\, P(w_j \mid ham)$$

$\alpha - 1000$ ✓ : $n_1 = 100$ (spam) $n_2 = 900$ (ham)

$$P(w_j \mid spam) = \frac{90 + 1000}{100 + 2 \cdot 1000} = \frac{1090}{2100} \sim 0.5$$

$$P(w_j \mid ham) = \frac{n_2' + \alpha}{n_2 + C \cdot \alpha} = \frac{100 + 1000}{900 + 2000}$$

$$= \frac{1100}{2900} \sim 0.4$$

~ 0.5

α - on The prob. underfit

		likelihood ~ 0.5	
$P(spam \mid w_j)$	$P(spam)$ ✓	$P(w_j \mid spam)$	$P(ham) \cdot P(w_j \mid ham)$ ✓
$\alpha = 1000$	$0.7 \times$	0.5	0.3×0.4
$P(spam \mid w_j)$	$P(spam) \times$	$P(w_j \mid spam)$	$P(ham) \times P(w_j \mid ham)$
$\alpha = 1$	0.7	0.9	$0.3 \quad 0.1$

large α - underfit - $P(spam \mid w_j) \simeq P(spam)$

likelihoods are ~ 0.5

Moderate value of $\alpha = 1$ or 5.

$P(spam \mid w_j)$ dependent on likelihoods.

Very small value of 'α' 0.1

$$P(w_j \mid spam) = \frac{n_1' + \alpha}{n_1 + C \cdot \alpha} = \frac{90 + 0.1}{100 + 2(0.1)}$$

$$\sim = \frac{90.1}{100.2} \quad 0.9$$

$$= \frac{90 + 0.01}{100 + 0.03}$$ high weightage

for likelihood based on the presence of words

0.5 ✓

0.9 . $\frac{90}{100}$ $\quad \frac{90 + 0.01}{100 + 0.03}$ vs $\frac{50 + 0.01}{100 + 0.03}$ $\frac{50}{100}$.

→ lead to overfitting. $\boxed{\text{"reply"}}$ Spam

α – hyper parameter controls overfitting in "NB".

Class Imbalance ?

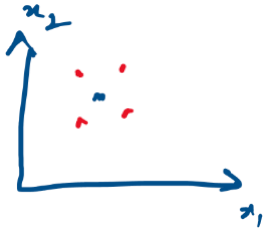
$P(\text{spam} \mid w_j)$ 100 $P(\text{ham} \mid w_j)$

$P(\text{spam}) \cdot P(w_j \mid \text{spam})$ $P(\text{ham}) \cdot P(w_j \mid \text{ham})$

0.2 0.4 ~ 0.08 0.8 0.1 ~ 0.08

$\frac{90}{900} = 0.1$

$P(w_j \mid \text{spam}) = \frac{40}{100}$ ✓



Imbalanced → Blance. ?

→ Text → I will take off tomorrow?

loss function – Class weight.

error (70) ham (30) spam. error × 3?

1

→ Oversampling | under Sampling ✓

ham	spam
400	300

→ Underflow problem a value of 10^{35} ?

Precision – Naive $j = 100$

$$P(\text{Spam} \mid w_1\ w_2 \ldots w_j) = P(\text{spam}) \cdot P(w_1 \mid s) \cdot P(w_2 \mid \text{spam}) \ldots P(w_j \mid \text{spam})$$

$$\Rightarrow [0.3 \times 0.1 \times 0.2 \times 0.3 \ldots 0.45]$$

$$\sim 0.0000000 \ldots 78$$

Underflow

Spam

$$= P(\text{Spam}) \cdot \prod_{j=1}^{d} P(w_j \mid \text{spam})$$

ham

$$P(\text{ham}) \cdot \prod_{j=1}^{d} P(w_j \mid \text{ham})$$

log =

$$\log(a_1 \cdot b_1 \cdot c_1 \cdot d_1 \ldots)$$

$$\log(a_2 b_2 c_2 \ldots)$$

$$\log a_1 + \log a_2 + \log a_3 + \ldots$$

$$0.1 \times 0.2 \times 0.3 \times 0.4 \times 0.9 \ldots ?$$

$$\log(0.1) + \log(0.2) + \log(0.3) \ldots$$

$$\uparrow 0.9$$

0.3

Feature Importance ?

likelihoods – as feature importance.

• Spam

w_i ↑ likelihood – 0.5

ham ↑

W_3 – likelihood – 0.9

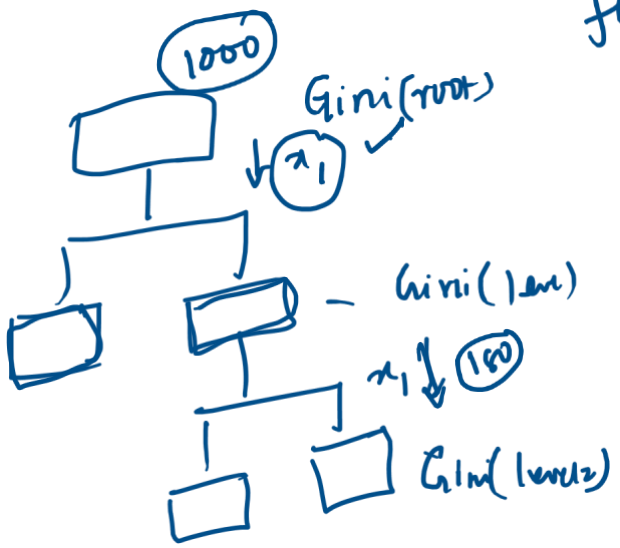
features (words)

$$P(\text{spam} \mid \text{word}) \sim \underbrace{P(\text{spam})}_{\text{fixed}} \cdot \underbrace{P(\text{word} \mid \text{spam})}_{\text{likelihood}}$$

$$P(\text{ham} \mid \text{word}) \sim P(\text{ham}) \cdot P(\text{word} \mid \text{ham})$$

feature importance → $\boxed{\text{Prediction}}$

(weighted) DT
Mean decrease in Gini



features of DT

Naive Bayes. likelihood ~ Prediction < Spam / ham

$p(\text{spam} \mid \text{word})$ $\qquad$ $p(\text{ham} \mid \text{word})$

0.9 $\quad p(\text{word} \mid \text{spam}) \uparrow$ $\qquad$ $p(\text{word} \mid \text{ham}) \uparrow$

→ Impact of outliers? a very rare word.

most frequent word?

Spam (word)	ham
$\frac{99}{100}$	$\frac{10}{900}$

word - Spam word - one document.

$$p(\text{word} \mid \text{spam}) = \frac{1}{100} - 0.01 \checkmark \quad \frac{1}{1000} \sim 0.00$$

$p(\text{spam} \mid w_1, w_2 \ldots w_n)$

$$\underbrace{P(\text{spam}) \cdot p(w_1 \mid \text{spam}) \ldots p(w_k \mid \text{spam})}$$

0.001

laplace Smoothing?

outlier w' — Not in training.

w' — testing.

$$P(w' \mid spam) \sim \frac{n_1'}{n_1} = \frac{0}{100} \rightarrow 0$$

$$P(w' \mid spam) \sim \frac{n_1' + \alpha}{n_1 + C \cdot \alpha} \sim \frac{0+1}{100+2} = \frac{1}{102} \rightarrow 0.01$$

→ **solution** :- ignore rare words ?

if the frequency of a word < 5 documents

Remove 'word'

→ $$P(spam \mid word) \sim P(spam) \cdot P(word \mid spam)$$?

$$P(word \mid spam) = \frac{90}{100} \sim 0.9 \ ?$$

Multinomial Naive Bayes :-

$$P(\text{Spam} \mid \text{word}\{\underbrace{1, 2, 3, 4, 5\ 10}\})$$

likelihoods

$$P(\text{Spam}) \cdot P(\text{word}_1 \mid \text{Spam}) \checkmark$$

$$P(\text{Spam})\ P(\text{word}_2 \mid \text{Spam}) \checkmark$$

$$P(w \mid \text{Spam}) \cdot P(w \mid \text{ham})$$

Multinomial.

Multinomial likelihoods.

$$P(w'_{(1)} \mid \text{Spam}) \qquad P(\text{word}'_{(1)} \mid \text{ham})$$

$$P(w'_{(2)} \mid \text{Spam}) \qquad P(\text{word}'_{(2)} \mid \text{ham})$$

$$\vdots \qquad \vdots$$

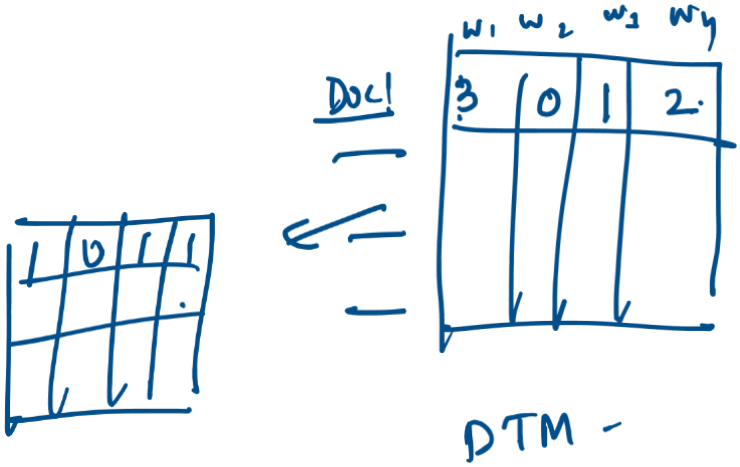
$$P(w'_{(7)} \mid \text{Spam}) \qquad P(\text{word}'_{(7)} \mid \text{ham})$$

Text. $\longrightarrow$ Document Term Matrix.

Collection of Text Documents
$\rightarrow$ Corpus

Binary feat

	w_1	w_2	...	w_m
1	0	1	1	0 ...
2				
⋮				
N				



Tf

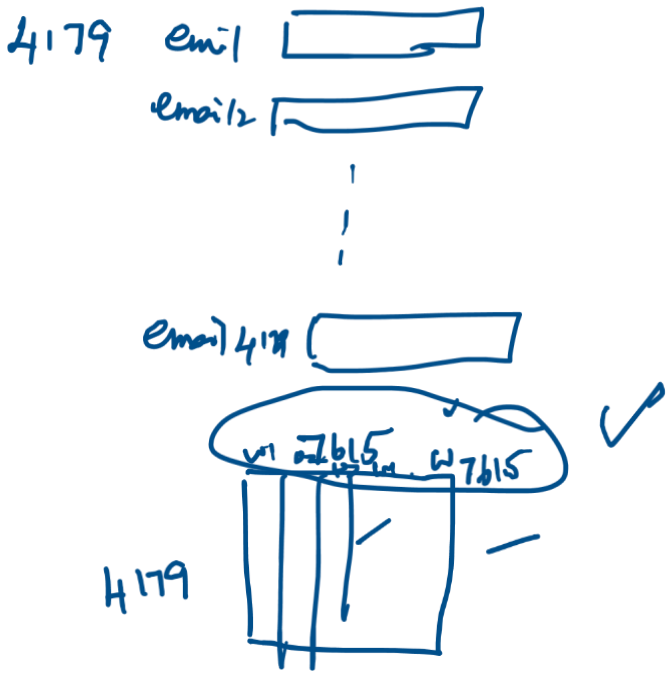
Tf

Multinomial Naive Bayes

Text → DTM (Matrix) → Vectorization

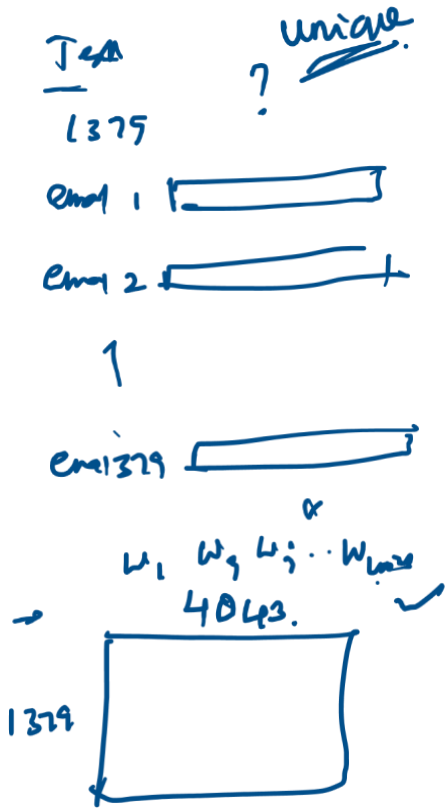
CountVectorizer() – Term frequencies

Train

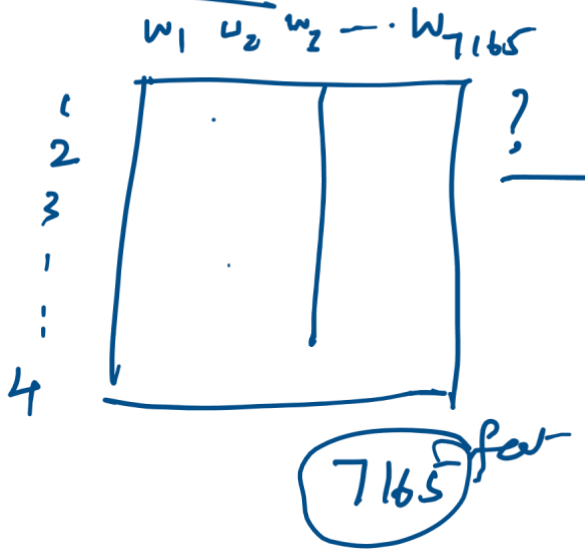


Test

1379

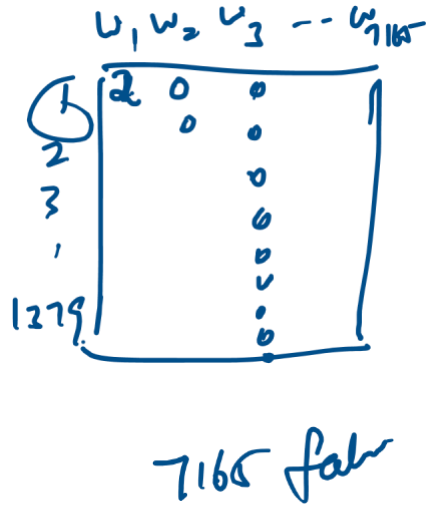


fit_transform(X_train)



X_train

transform(X_test)



(Y_train)